GEOMETRY · UNIT 4: TRIANGLE RELATIONSHIPS

Emergency Services Locator

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Three towns must share new emergency services. Students use perpendicular bisectors, medians, and angle bisectors to locate the circumcenter, centroid, and incenter, apply the Triangle Inequality and midsegments to road plans, and recommend locations for a different set of towns.

Each point of concurrency answers a different question. Using a right triangle makes the circumcenter land on the hypotenuse, which surprises students and leads to a good discussion.

Standards Addressed

- G-CO.10 Prove theorems about triangles (medians, midsegments)
- G-C.3 Construct inscribed and circumscribed circles
- G-GPE.4 Use coordinates to prove theorems
- G-CO.12 Make formal constructions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: where would you put a hospital? Part 1 circumcenter and centroid.
Day 2	Part 2 Triangle Inequality, midsegments, and angle order.
Day 3	Part 3 new towns and proposal.

Materials

- Student handout
- Compass, straightedge, and protractor
- Graph paper
- Optional: patty paper for folding bisectors

How the Three Versions Differ

Level	What changes
CP2	Midpoints and averages for the circumcenter of a right triangle and the centroid; Triangle Inequality and midsegment facts.
CP1	Equations of perpendicular bisectors, verifying equidistance, medians and the centroid's 2 : 1 ratio, and ordering angles.
Honors	Incenter and inradius, the Euler line, and a proposal for a non-right triangle.

Watch For (Common Misconceptions)

- Mixing up circumcenter (towns) and incenter (roads).
- Thinking the circumcenter is always inside the triangle.
- Forgetting the Triangle Inequality must hold for all three pairs.

Extension Ideas

- Construct the points with a compass and verify.
- Use GeoGebra to drag the towns and watch the points move.
- Research how real hospitals are sited.

Answer Key – CP2

Part 1: Hospital & Warehouse

#	Task	Answer
1	Classify triangle ABC by its sides and by its angles.	Scalene right triangle (right angle at Ashford).
2	The hospital must be equidistant from all three towns (the circumcenter). In a right triangle, it is the midpoint of the hypotenuse. Find it.	Midpoint of BC: (6, 3)
3	Check: find the distance from the hospital to each town.	Each distance is $\sqrt{(6^2 + 3^2)} = \sqrt{45} \approx 6.71$ miles.
4	The warehouse goes at the centroid: average the x-coordinates and the y-coordinates of the towns. Find it.	$((0 + 12 + 0) \div 3, (0 + 0 + 6) \div 3) = (4, 2)$

Part 2: Roads & Bridges

#	Task	Answer
5	Could three roads of 5, 8, and 14 miles form a triangle? Explain.	No: $5 + 8 = 13 < 14$.
6	Two roads are 7 and 12 miles long. Write an inequality for the possible length x of the third road.	$12 - 7 < x < 12 + 7 \rightarrow 5 < x < 19$
7	A bridge connects the midpoints of AC and BC. Find the midpoints and the length of the bridge. How does it compare to AB?	Midpoints (0, 3) and (6, 3); length 6 miles, half of AB (12), and parallel to AB.
8	List the angles of triangle ABC from smallest to largest. Explain using the side lengths.	Sides: AC = 6, AB = 12, BC = $\sqrt{180} \approx 13.4$. Smallest angle is opposite the shortest side: $\angle B$, then $\angle C$, then $\angle A$ (90°).

Part 3: Your Proposal

#	Task	Answer
9	Find the hospital (circumcenter) and the warehouse (centroid) for the new towns.	Circumcenter = midpoint of QR = (4, 5); centroid = $(8 \div 3, 10 \div 3) \approx (2.67, 3.33)$
10	Show that the hospital is the same distance from all three new towns.	Each distance is $\sqrt{(4^2 + 5^2)} = \sqrt{41} \approx 6.40$ miles.
11	Write a recommendation to the county with a labeled map.	Answers vary.

Answer Key – CP1

Part 1: Hospital & Warehouse

#	Task	Answer
1	Write the equations of the perpendicular bisectors of AB, AC, and BC.	AB: $x = 6$. AC: $y = 3$. BC: midpoint (6, 3), slope of BC is $-\frac{1}{2}$, so $y - 3 = 2(x - 6) \rightarrow y = 2x - 9$.
2	Find where they meet and verify it is equidistant from all three towns.	All meet at (6, 3); each town is $\sqrt{45} \approx 6.71$ miles away.
3	Find the centroid. Then show it is $\frac{2}{3}$ of the way from A to the midpoint of BC.	Centroid (4, 2). Midpoint of BC is M(6, 3). $A + \frac{2}{3}(M - A) = (4, 2)$ ✓

Part 2: Roads & Bridges

#	Task	Answer
4	Two roads from Ashford are 6 and 12 miles. What lengths are possible for a road that closes the triangle? Is BC's length in this range?	$6 < x < 18$. BC = $\sqrt{180} \approx 13.4$, which is in the range.
5	A bridge joins the midpoints of AC and BC. Prove it is parallel to AB and half as long, using slopes and distance.	Midpoints (0, 3) and (6, 3): slope 0, same as AB, so parallel; length $6 = \frac{1}{2} \cdot 12$.
6	Order the angles of triangle ABC from smallest to largest and the sides from shortest to longest. Explain the connection.	$\angle B < \angle C < \angle A$; AC (6) < AB (12) < BC (≈ 13.4). The largest angle is opposite the longest side.

Part 3: Your Proposal

#	Task	Answer
7	Find the circumcenter using perpendicular bisectors.	PQ bisector: $x = 5$. PR: midpoint (2, 3), slope of PR = 1.5, perpendicular slope $-\frac{2}{3}$: $y - 3 = -\frac{2}{3}(x - 2)$. At $x = 5$: $y = 1$. Circumcenter (5, 1).
8	Find the centroid. Is the circumcenter inside, on, or outside the triangle? Why?	Centroid $(14 \div 3, 2) \approx (4.67, 2)$. The circumcenter (5, 1) is inside: the triangle is acute (all angles less than 90°).
9	Write a recommendation with a labeled map.	Answers vary.

Answer Key – Honors

Part 1: Three Centers

#	Task	Answer
1	Find the circumcenter and centroid. Justify each.	Circumcenter (6, 3) (midpoint of the hypotenuse); centroid (4, 2) (average of vertices).
2	The fire station must be equidistant from the three roads (the sides). For a right triangle with legs a and b and hypotenuse c, the inradius is r...	$r = (6 + 12 - \sqrt{180}) \div 2 = 9 - 3\sqrt{5} \approx 2.29$; incenter $(9 - 3\sqrt{5}, 9 - 3\sqrt{5}) \approx (2.29, 2.29)$
3	Find the orthocenter. Then show that the orthocenter, centroid, and circumcenter are collinear (the Euler line).	Orthocenter A(0, 0) (the legs are altitudes). (0, 0), (4, 2), (6, 3) all lie on $y = \frac{1}{2}x$, so they are collinear.

Part 2: Proofs & Roads

#	Task	Answer
4	Prove that any point on the perpendicular bisector of AB is equidistant from A and B.	If P is on the bisector, P and the midpoint M give right triangles PMA and PMB with AM ≅ BM and PM ≅ PM, so △PMA ≅ △PMB (SAS) and PA ≅ PB (CPCTC).
5	Prove that the midsegment joining the midpoints of AC and BC is parallel to AB and half its length, using general coordinates A(0, 0), B(2a, 0),...	Midpoints (b, c) and (a + b, c): slope 0 = slope of AB; length a = ½(2a).
6	A new road from Cedar to a point D on AB must be shorter than CB and longer than CA. Describe all possible locations of D.	CD must satisfy $6 < CD < \sqrt{180}$. With D = (d, 0), $CD = \sqrt{d^2 + 36}$): so $0 < d < 12$: every point strictly between A and B works.

Part 3: Proposal for an Obtuse Region

#	Task	Answer
7	Find the circumcenter and explain why it is outside the triangle.	PQ bisector x = 5; PR: midpoint (1, 1.5), slope 1.5, perpendicular $-\frac{2}{3}$: $y - 1.5 = -\frac{2}{3}(x - 1) \rightarrow$ at x = 5, $y = 1.5 - 8 \div 3 \approx -1.17$. Circumcenter $(5, -7 \div 6) \approx (5, -1.17)$. Outside because ∠R is obtuse.
8	Argue whether the circumcenter is still a good hospital location. Propose an alternative and justify it with distances.	Answers vary.
9	Write the full proposal with a map.	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Locations are correct for the stated goal (equidistant from towns, balance point, equidistant from roads). Goes beyond the minimum with a creative, realistic choice.	Locations are correct for the stated goal (equidistant from towns, balance point, equidistant from roads).	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Proposal shows each point of concurrency on a map with calculations and a recommendation. Polished, organized, and visually engaging.	Proposal shows each point of concurrency on a map with calculations and a recommendation. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.